

Can a Vision-Language Model Group 2D Masks into 3D Objects?

VLM-based edge scoring and geometric-semantic fusion for 3D instance segmentation

Samet Degirmenci · Florian K. · Katerina Skotnicova · Omar A.

Machine Learning for 3D Geometry (SS 2026) | Technical University of Munich



Code

TUM

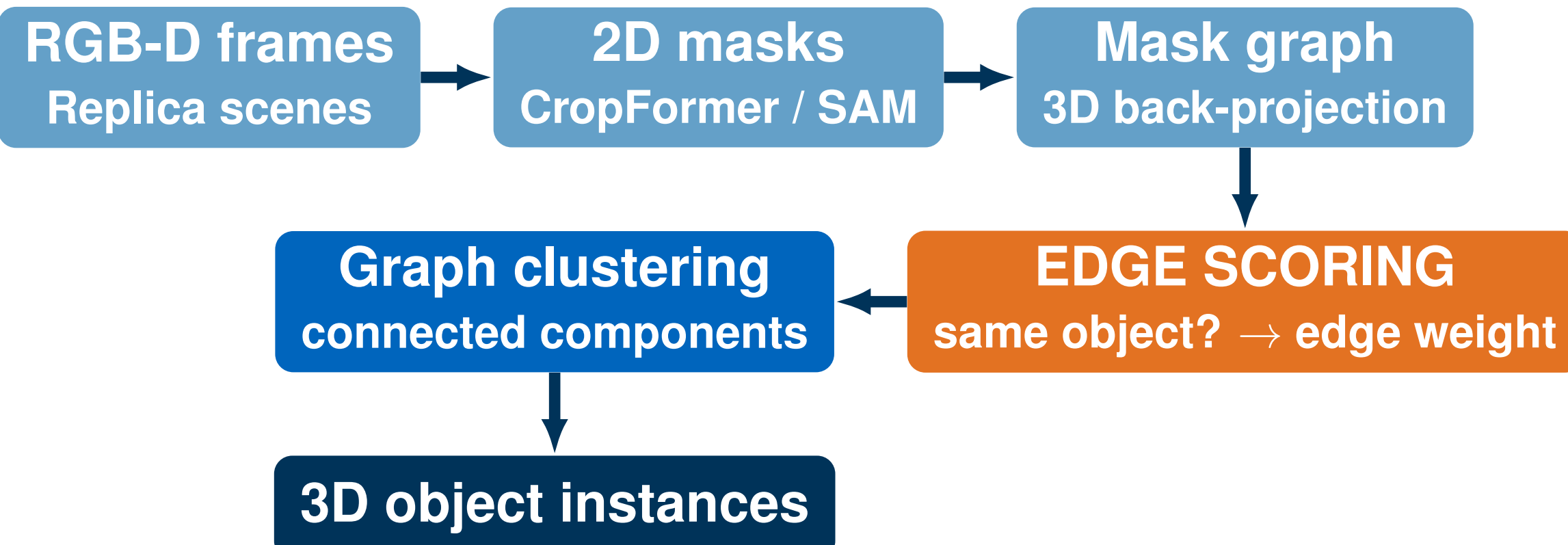
Motivation



Class-agnostic 2D masks (CropFormer) in one RGB-D frame of Replica room0.

- Goal: group per-frame **2D instance masks** into consistent **3D objects**.
- MaskClustering (CVPR'24) solves this **purely geometrically** — no semantics.
- Can a vision-language model (**Qwen3-VL**) judge “same object?” instead — or in addition?
- **We ablate two axes**: the 2D mask backbone (CropFormer vs SAM 1/SAM 3) and the edge scorer (geometry → VLM → **learned fusion**).

Method: One Pipeline, Three Edge Scorers



Masks become **graph nodes**; candidate pairs (3D overlap $\geq \tau$) get an edge if the **edge score** passes a threshold. Connected components form 3D objects. **We keep the pipeline fixed and swap only the edge scorer.**

Edge Scoring: Three Approaches

1 · GEOMETRIC (baseline)

View consensus: a mask pair is merged if enough frames that observe both also contain them inside one mask (*supporters* / *observers*). Purely geometric — fails on adjacent objects with high 3D overlap.

2 · SEMANTIC (Qwen3-VL)



What the VLM sees: two views, one region outlined each (A red, B blue).

The VLM answers “**same physical object?**” with a confidence score, used directly as the edge weight. **Zero-shot** it is over-conservative (too few edges, fragmented scenes); **LoRA finetuning** on 5 held-out Replica scenes sharpens it, but raw VLM edges stay too sparse to group instances alone.

3 · LEARNED FUSION (MLP)

A small MLP (2×32 , $\sim 1.3k$ params) combines **geometric features** (view consensus, #observers, 3D IoU) with the **Qwen confidence** into one edge probability. Trained on 6 scenes, validated on 1, tested on *room0*.

Question: does semantics add anything on top of geometry? **On Replica, barely — geometry dominates.**

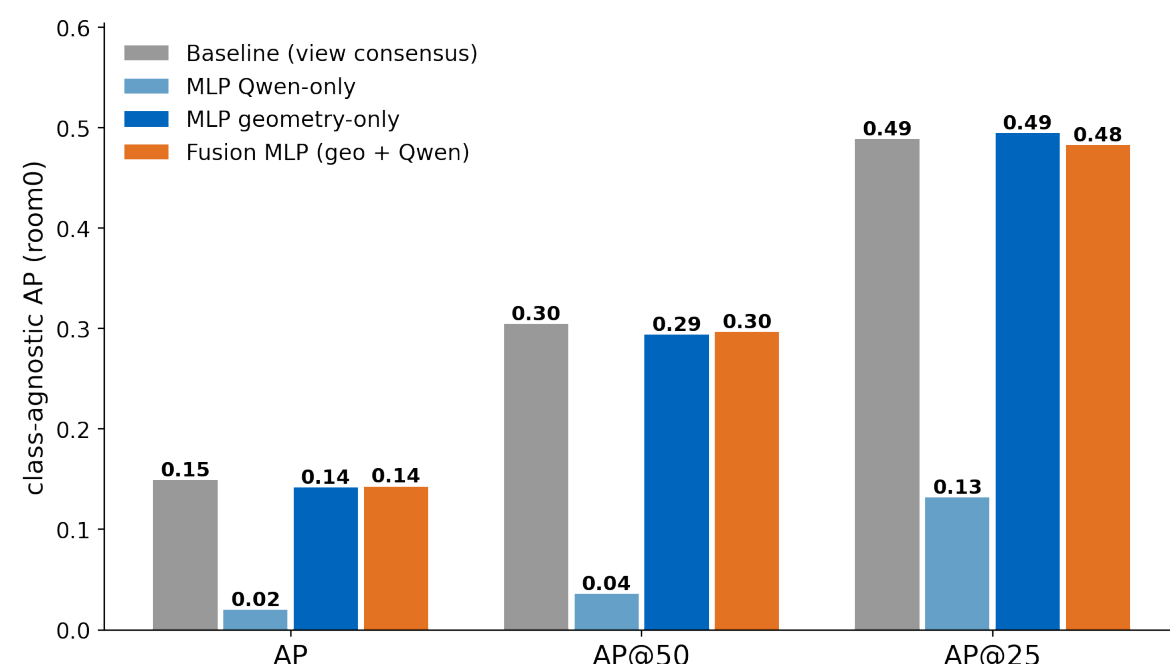
Summary & Outlook

- **CropFormer is the strongest mask backbone (M2)**: it beats SAM 1 and SAM 3 on downstream 3D AP, so all later milestones build on CropFormer masks.
- **The raw VLM signal is weak**: zero-shot and finetuned Qwen edges give AP ≈ 0.02 — too sparse to group 3D instances on their own.
- **Geometry drives the fusion MLP**: it reaches baseline-level AP, but adding the Qwen score moves AP by < 0.01 — semantics adds little on top of geometry on Replica.

KEY FINDING The fusion MLP matches the geometric baseline but does not surpass it: on Replica the VLM contributes almost no signal once view-consensus geometry is in the model.

NEXT STEPS Stronger VLM supervision · projection-based single-image prompting · more scenes and larger VLMs.

Results — Replica room0 (class-agnostic AP)



Edge scorer	AP	AP@50	AP@25
View consensus (baseline)	0.149	0.305	0.489
Qwen finetuned (LoRA)	0.020	0.027	0.150
MLP Qwen-only	0.020	0.036	0.132
MLP geometry-only	0.142	0.294	0.495
Fusion MLP (geo + Qwen)	0.143	0.297	0.483

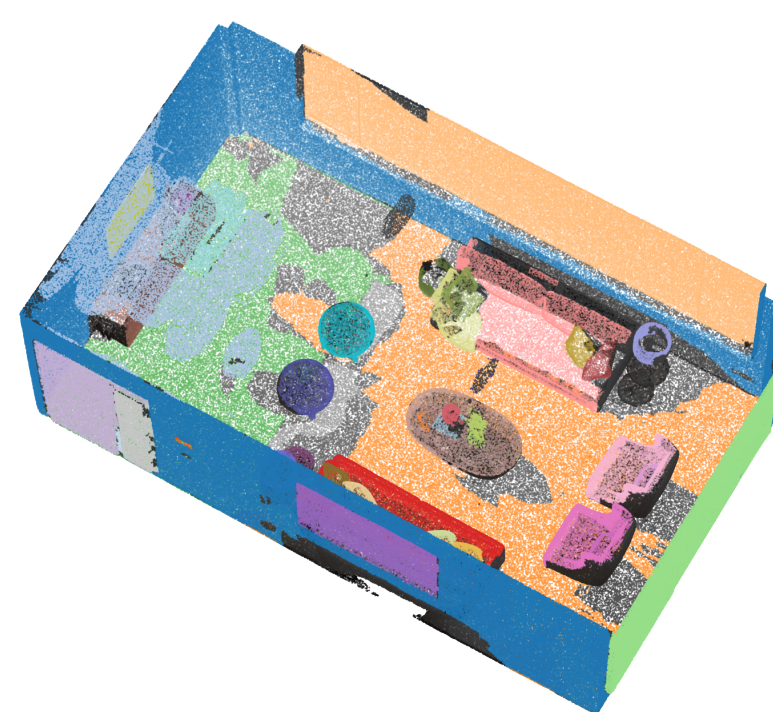
The fusion MLP matches the baseline (AP 0.143 vs 0.149) but does not beat it; Qwen-only edges stay near zero. All variants share the same clustering and CropFormer masks.

Mask backbone (M2)	AP	AP@50	AP@25
CropFormer	0.149	0.305	0.489
SAM 3 (conf 0.25)	0.123	0.224	0.370
SAM 1 (ViT-H)	0.061	0.107	0.222

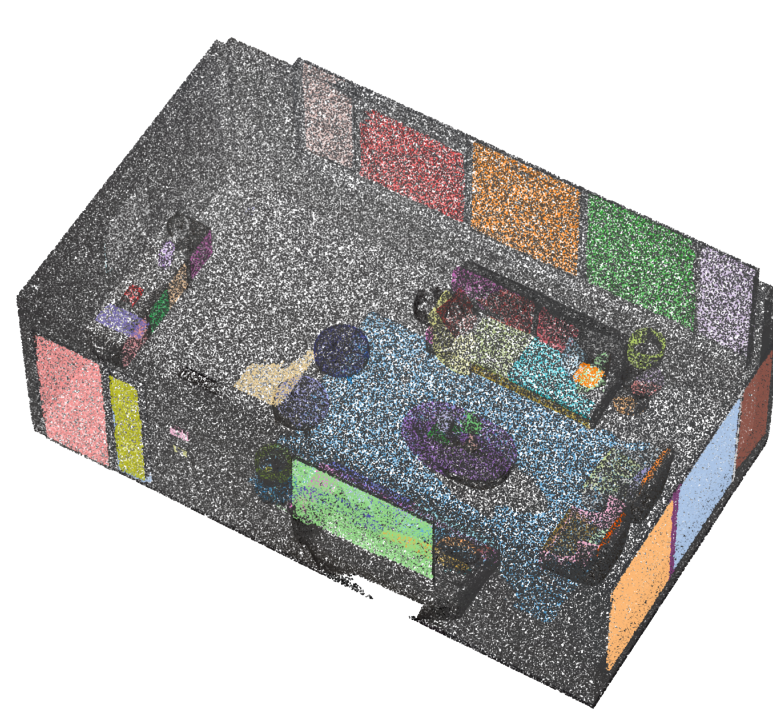
Same geometric pipeline, different 2D masks. CropFormer's class-agnostic masks give the best 3D AP, so all later milestones use them.

Predicted 3D Instances

Baseline (CropFormer)



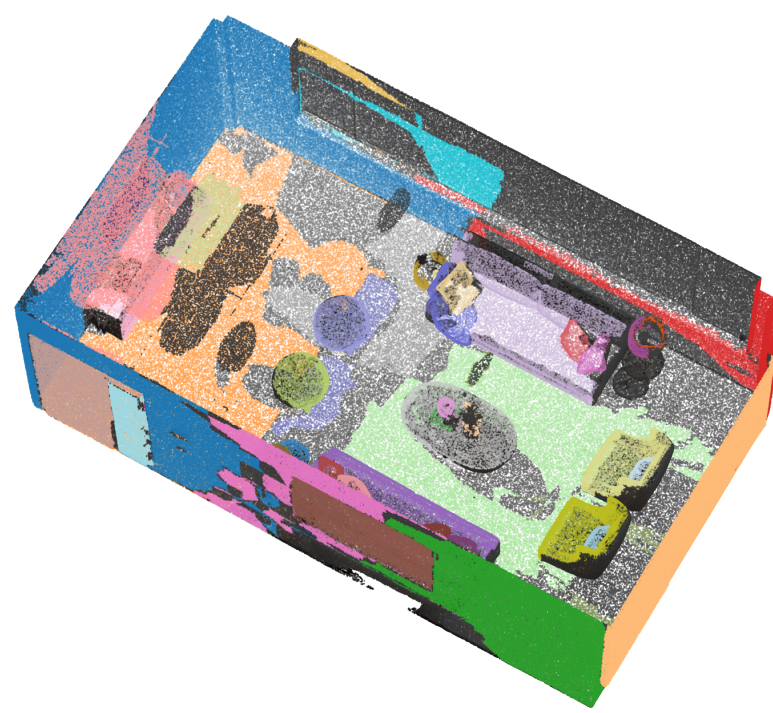
SAM 3 masks



Qwen finetuned



Fusion MLP



One color per predicted instance, same viewpoint. SAM 3 masks over-segment surfaces (walls, floor) into large slabs; CropFormer + geometry recover cleaner objects, and the fused MLP predicts 100 instances — closest to the true count.